



Extra Activity - 2



The due date for submitting this assignment has passed.

Due on 2026-03-30, 23:59 IST.

You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-03-20, 08:51 IST

In this activity, we will take a look at the application of geogebra to visualize linear transformations $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$. The notation in the questions that follow will be as in the following project

<https://www.geogebra.org/m/krqstvdm>

The description of the elements in the project is as follows:

- The parameters a, b, c and d are variables used to determine the linear transformation whose matrix representation with respect to the standard bases is $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$.
- $T\mathbf{e}_1$ and $T\mathbf{e}_2$ are vectors which represent the images of the standard basis vectors $\mathbf{e}_1$ and $\mathbf{e}_2$, respectively.
- $\mathbf{u}$ is an arbitrary vector in the plane, and $T\mathbf{u}$ is its image under T .
- P is a quadrilateral in the plane and TP is its image under T .
- S is a circular sector in the plane and TS represents its image under T .

To specify the transformation, one can specify the values of a, b, c and d . Or alternatively, move the vectors $T\mathbf{e}_1$ and $T\mathbf{e}_2$ to specify the image of the standard basis, thus determining the linear transformation. Once a linear transformation is fixed, the vector $\mathbf{u}$ can be moved by dragging its arrowhead. The vector $T\mathbf{u}$ changes accordingly to give the image of the vector $\mathbf{u}$. Similarly, the elements P and S can be moved around and TP and TS can be analyzed to understand the geometric properties of the linear transformation. Answer the following questions using the given project.

1 point

Consider the linear transformation T which scales each of the standard basis vectors by a non-zero constant. In the geogebra project, this corresponds to moving the vectors $T\mathbf{e}_1$ and $T\mathbf{e}_2$ so that the arrowheads are on $(\lambda, 0)$ and $(0, \lambda)$, respectively, for some $\lambda \in \mathbb{R} \setminus \{0\}$ of your choice.

Which of the following statements are true for such a linear transformation?

- ☒ The matrix of T with respect to the standard bases for the domain and codomain is a scalar matrix.

If $\lambda < 0$, the angle between $\mathbf{u}$ and $T(\mathbf{u})$ is 180° , for all $\mathbf{u} \in \mathbb{R}^2 \setminus \{(0, 0)\}$.

☐ If P is the square with vertices $(0, 0), (1, 0), (1, 1)$ and $(0, 1)$, then its image TP has area equal to λ square units.

☒ The linear transformation T is always a bijection.

☐ If $\lambda > 0$, then there is a non-zero vector $\mathbf{u}$ such that $T(\mathbf{u}) = \mathbf{u}$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

The matrix of T with respect to the standard bases for the domain and codomain is a scalar matrix.

If $\lambda < 0$, the angle between $\mathbf{u}$ and $T(\mathbf{u})$ is 180° , for all $\mathbf{u} \in \mathbb{R}^2 \setminus \{(0, 0)\}$.

The linear transformation T is always a bijection.



It is known that the matrix of a linear transformation T which rotates every vector counter-clockwise by an angle of θ is given by

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

In the geogebra project, set

$$a = 0.6,$$

$$b = -0.8,$$

$$c = 0.8,$$

$$d = 0.6$$

This gives a transformation which rotates every vector in the plane. Consider the sector SS whose vertices are $(0, 0), (1, 2)$ and $(2, 1)$. Under the above transformation, the image of SS is a sector $TSTS$, which is rotated counter-clockwise by an angle $\sin^{-1}(0.8)$. If the vertices of $TSTS$ are $(0, 0), (-1, 2)$ and $(0.4, k)$, find the value of k .

2.2

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 2.2

1 point

Move the head of the vector $T e_1$ to $(1, -1)$. Observe the image $T P$ of the polygon P as you vary the vector $T e_2$ such that its head lies on the line $x = -2$. There is exactly one point $(-2, y_0)$ such that $T P$ is a line segment. For all other points, the image of P is a quadrilateral. Find the value of y_0 .

2

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 2

1 point

1 point

Consider the transformation in the previous question for which the image of P is a line segment. Which of the following statements are true about the transformation?

- ☒ The image of the transformation is a line through the origin.
- ☐ The set $\{(1, 1)\}$ is a basis for the Kernel of the transformation.
- ☐ The matrix of the transformation is invertible.
- ☒ There is exactly one line through the origin whose image under the transformation is the singleton set $\{(0, 0)\}$.
- ☒ The set $\{(-1, 1)\}$ is a basis for the image of the transformation.

Yes, the answer is correct.

Score: 1

Accepted Answers:

The image of the transformation is a line through the origin.

There is exactly one line through the origin whose image under the transformation is the singleton set $\{(0, 0)\}$.

The set $\{(-1, 1)\}$ is a basis for the image of the transformation.